PROJECT MANAGEMENT & OPERATIONAL PLAN

AI-Powered Support Ticket Classifier

Human-in-the-Loop Review System Implementation Specification

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PROJECT TEAM & RESOURCE ALLOCATION

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1 EXECUTIVE SUMMARY & OVERVIEW

Customer support operations in modern organizations face a persistent operational bottleneck: thousands of incoming support tickets must be manually read, categorized, and routed to the correct department. Industry estimates indicate that support teams waste **30–40% of operational capacity** on manual triage rather than resolving customer issues.

Current automated solutions fail by adopting rigid extremes: fully manual routing is slow and error-prone, while fully automated ML routing poses silent risks when handling ambiguous inputs. This project delivers an operational ML pipeline combining classical NLP (TF-IDF + Logistic Regression) with a **Human-in-the-Loop (HITL)** review mechanism for balanced efficiency and precision.

> **Core System Workflow:** 1. Classifies tickets automatically | 2. Assigns probability Confidence Scores | 3. Escalates low-confidence tickets (<70%) to a Review Queue | 4. Logs all decisions in an Audit Log for feedback.

2 PROJECT PROBLEM & CORE CAPABILITIES

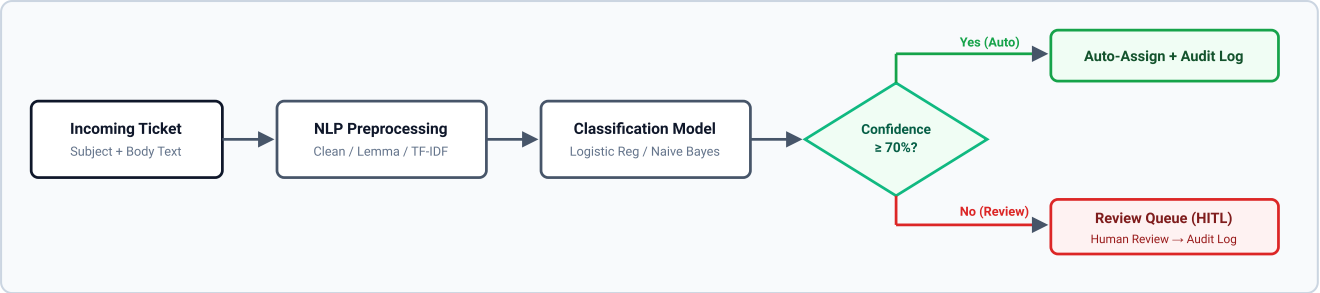
2.1 Key Operational Challenges

- **Manual Triage Bottleneck:** High ticket volume creates response delays.
- **Automation Risk:** Unverified models produce silent misclassifications.
- **Missing Feedback Loop:** Manual corrections are rarely logged for retraining.
- **Human Oversight Gap:** Autonomous AI lacks escalation mechanisms.

2.2 Technical Solution Delivery

- **NLP Classification:** NLTK/spaCy text cleaning + TF-IDF representation.
- **Confidence Thresholds:** `predict_proba()` scoring with 70% threshold.
- **Streamlit HITL Dashboard:** Real-time Review Queue for human validation.
- **Traceable Audit Log:** SQLite/CSV record of reviewer actions & timestamps.

3 SYSTEM WORKFLOW ARCHITECTURE



4 PROJECT PLANNING & MANAGEMENT

4.1 Project Timeline (5-Week Sprint Roadmap)

SPRINT PHASE	PHASE NAME	KEY OPERATIONAL DELIVERABLES
`Week 1`	**Data Prep & Baseline**	Dataset collection, text preprocessing (NLTK/spaCy), TF-IDF vectorization, baseline model training.
`Week 2`	**Model & Confidence**	Model tuning, evaluation (Accuracy, F1-score), implementing `predict_proba()` confidence extraction.
`Week 3`	**HITL & Audit Database**	Setting routing threshold (70%), building Review Queue logic, designing SQLite/CSV Audit Log.
`Week 4`	**Dashboard & Integration**	Developing Streamlit dashboard (Auto-classification + Review Queue), experiment tracking with MLflow.
`Week 5`	**Testing & Delivery**	End-to-end integration testing, error analysis, final report writing, live demo & presentation preparation.

4.2 Team Roles & Responsibilities

TEAM MEMBER	ASSIGNED ROLE	CORE RESPONSIBILITIES
Member 1	NLP & Data Engineer	Dataset acquisition, text preprocessing (NLTK/spaCy), TF-IDF vectorization, data quality assurance.
Member 2	Machine Learning Engineer	Model training, hyperparameter tuning, class imbalance handling, evaluation report generation.
Member 3	Backend & Integration	Confidence scoring logic, threshold routing implementation, Audit Log database architecture.
Member 4	Frontend Developer	Streamlit dashboard UI development (Auto-classification view + Review Queue interface).
Member 5	MLOps & Doc Lead	MLflow experiment tracking setup, code documentation, final report writing, live presentation lead.

4.3 Risks & Mitigation Strategies

IDENTIFIED RISK	PROB.	IMPACT	MITIGATION STRATEGY
Time Constraint (5 Wks)	HIGH	HIGH	Adhere strictly to MVP scope; utilize Streamlit for rapid UI development without custom JavaScript.
Dataset Class Imbalance	MEDIUM	MEDIUM	Apply class weights in loss function, or use resampling techniques (SMOTE) during model training.
Threshold Tuning Issue	MEDIUM	MEDIUM	Experiment with confidence thresholds (60%, 70%, 80%) to optimize auto-classification rate vs accuracy.
Team Unavailability	LOW	HIGH	Cross-training team members for shared ownership of core pipeline modules.

4.4 Key Performance Indicators (KPIs)

≥ 85%	≥ 75%	20-25%	< 1.0s	100%
MODEL ACCURACY	AUTO-CLASSIFY RATE	QUEUE ROUTING	SYSTEM LATENCY	AUDIT LOGGING

5 REFERENCES

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